

MTHATHA, South Africa

WMO: 686680

Lat: 31.5497S

Lon: 28.6739E

Elev: 747

StdP: 92.67

Time Zone: 2.00 (E02)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.0	4.1	-7.1	2.3	18.8	-5.0	2.7	18.4	11.2	14.4	9.7	16.4	1.0	270	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.1	34.5	19.8	32.8	19.6	31.0	19.4	23.3	29.7	22.4	28.7	21.6	27.6	3.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	17.6	26.5	20.5	16.6	25.5	19.8	15.9	24.6	73.4	29.7	69.9	28.7	66.8	27.8	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	7.9	6.9		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
				-0.1	25.2	0.6	0.7	-0.6	25.8	-0.9	26.2	-1.2	26.6	-1.6	27.1
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	22.0	22.3	21.1	18.7	16.6	13.5	13.4	15.3	16.8	18.1	19.1	20.5
	DBStd	4.39	3.01	2.94	3.18	3.36	3.07	2.62	2.89	3.40	3.73	3.77	3.70	3.24
	HDD10.0	11	0	0	0	0	1	3	5	2	1	0	0	0
	HDD18.3	704	6	3	11	36	70	146	154	107	72	51	34	15
	CDD10.0	2967	372	344	344	261	206	109	111	166	204	252	272	327
	CDD18.3	619	119	114	97	47	17	1	2	12	26	45	56	83
	CDH23.3	5031	772	737	647	396	220	58	88	232	361	449	472	599
	CDH26.7	1873	295	278	249	134	49	6	11	74	147	186	202	242
Wind	WSAvg	2.8	3.2	2.9	2.6	2.2	2.0	2.1	2.2	2.7	3.0	3.2	3.5	3.4
Precipitation	PrecAvg	634	88	91	98	51	20	13	14	20	29	67	77	86
	PrecMax	920	163	197	262	244	103	90	80	98	159	273	206	220
	PrecMin	378	11	13	13	11	1	0	0	0	0	11	3	15
	PrecStd	135	40	47	60	43	24	20	17	20	28	48	40	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	35.8	35.1	33.0	30.2	27.3	28.1	32.2	34.0	35.1	36.0	35.6
		MCWB	21.3	21.8	20.9	19.0	16.1	13.3	12.6	14.6	15.5	17.6	17.8	21.1
	2%	DB	32.9	32.5	32.2	30.4	28.1	25.1	25.9	28.8	30.8	32.0	32.1	32.2
		MCWB	21.4	21.9	20.7	18.3	15.0	12.3	12.0	13.3	14.9	17.2	19.6	20.7
	5%	DB	30.4	29.9	29.8	28.0	26.0	23.0	23.7	26.0	27.7	28.8	28.7	29.4
		MCWB	21.6	21.8	20.4	17.4	14.5	11.6	11.5	12.8	14.8	17.4	19.2	20.6
	10%	DB	28.0	27.7	27.2	25.4	23.5	20.8	21.1	22.9	24.2	25.7	25.6	26.7
		MCWB	21.0	21.2	20.1	17.4	14.4	11.0	10.8	12.1	14.4	17.1	18.1	19.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	24.7	24.4	21.6	19.6	14.7	14.7	16.5	18.5	20.8	22.1	23.5
		MCDB	31.4	30.9	29.9	27.1	23.8	22.7	22.1	23.7	26.9	28.6	29.2	30.4
	2%	WB	22.8	23.2	22.5	20.0	17.0	13.6	13.3	15.4	17.1	19.2	20.7	22.1
		MCDB	29.9	29.1	28.8	26.7	22.5	21.3	20.6	22.7	24.9	26.7	27.6	29.0
	5%	WB	22.1	22.2	21.3	19.0	16.0	12.8	12.5	14.5	16.2	18.3	19.7	21.1
		MCDB	28.5	27.9	26.8	25.4	22.4	20.0	20.0	21.5	23.4	25.6	26.3	27.6
	10%	WB	21.2	21.4	20.4	18.2	15.1	12.0	11.7	13.5	15.3	17.3	18.6	20.2
		MCDB	26.8	26.8	25.4	23.6	21.4	18.3	19.2	19.8	21.6	24.0	24.3	25.6
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.1	10.4	11.9	13.4	14.4	15.2	14.3	13.4	11.9	10.8	10.5
		MCDBR	14.8	14.3	15.2	16.1	17.2	18.3	18.9	19.2	19.3	18.1	16.8	16.3
		MCWBR	6.1	6.2	6.3	6.1	6.4	7.4	7.4	7.0	6.7	6.8	6.3	6.7
	5% WB	MCDBR	13.0	12.6	12.8	13.4	13.5	15.2	16.0	14.8	14.7	14.3	13.7	13.7
		MCWBR	6.0	6.1	6.0	5.8	6.2	6.9	7.3	6.4	6.4	6.4	6.3	6.5
Clear-Sky Solar Irradiance	taub		0.385	0.387	0.369	0.335	0.312	0.293	0.299	0.357	0.397	0.402	0.378	0.383
	taud		2.393	2.392	2.414	2.469	2.497	2.514	2.485	2.331	2.227	2.262	2.372	2.392
	Ebn at Noon		954	933	915	895	868	864	873	856	870	907	955	960
	Edh at Noon		127	124	113	96	83	77	83	108	134	139	129	128
All-Sky Solar Radiation	RadAvg		6.16	5.71	4.99	4.17	3.46	3.08	3.35	4.00	4.86	5.39	6.01	6.14
	RadStd		0.45	0.48	0.34	0.24	0.18	0.20	0.20	0.25	0.32	0.48	0.42	0.40

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (6 neighbors)	+0.24	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page